

Practicum No.14**Date: 20/3/2026****Risk & Insurance Planning for a CSE/AIML/DSCS/DS Enterprise****I. Practical Significance**

Risk & Insurance Planning is critical for technology-driven enterprises in CSE, AIML, DSCS, and Data Science, where rapid innovation also introduces significant vulnerabilities.

This practicum helps learners understand and map risks arising from cyber threats, AI model failures, data breaches, system downtime, cloud dependency, and regulatory non-compliance.

Students learn to evaluate technical, financial, operational, and legal risks that directly affect business continuity and investor confidence.

The practicum builds the ability to design structured risk mitigation plans tailored to AI/ML and data-centric environments. Learners gain insights into selecting suitable insurance products such as cyber liability insurance, Tech E&O, professional indemnity, and data breach insurance.

They also learn to classify risks into preventable, strategic, and external categories to strengthen enterprise resilience.

By applying risk heat maps, compliance frameworks, and incident response plans, learners develop systematic decision-making skills.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a Pitch deck, project proposals for small/medium-scale startups logically and scientifically.
- a) **Apply** analytical reasoning to identify, evaluate, and prioritize technical, financial, and cybersecurity risks affecting technology-driven enterprise.
- b) **Interpret** and assess different categories of risk (preventable, strategic, external) and map them to appropriate mitigation or insurance strategies.
- c) **Conduct** systematic risk analysis using tools such as risk heat maps, compliance checklists, and incident-response frameworks to generate evidence-based risk reports.
- d) **Practice** ethical standards, confidentiality, and integrity when handling sensitive data, risk information, and insurance-related documentation.

Relevant CO(s)

- **CO5. Develop** a pitch deck of the identified product/service and its management strategies for ROI.

III. Practical Outcome (PrO)

- **PrO14– Present** information on relevant insurance schemes covering risks for **the identified** CSE, AIML, DSCS & DS enterprise.

IV. Related ADO(s)

- a) **Communicate** effectively while presenting risk assessments, insurance options, and mitigation strategies (ADO).
- b) **Demonstrate** critical thinking and analytical curiosity in exploring complex technical, cybersecurity, and operational risks. (CDO).
- c) **Exhibit** ethical responsibility, confidentiality, and integrity when handling sensitive organizational risk data and insurance information. (ADO).

V. Minimum Underpinning Theory

Risk and Insurance Planning is a systematic process through which an organization identifies vulnerabilities, evaluates potential impacts, and implements structured strategies to protect assets, technology systems, and business continuity. In CSE, AIML, DSCS, and Data Science enterprises, risks often arise from cybersecurity threats, algorithmic failures, data breaches, cloud dependency, regulatory non-compliance, and ethical concerns related to AI-driven solutions. Understanding these risk factors enables technopreneurs to plan proactively and ensure sustainable operations in high-velocity digital environments.

Key theoretical concepts include:

- **Risk Identification:** Recognizing internal and external threats such as cyberattacks, model inaccuracies, server downtime, data privacy breaches, and intellectual property risks.
- **Risk Assessment:** Evaluating the likelihood and impact of each risk using tools like risk matrices, heat maps, and incident probability analysis.
- **Risk Mitigation Strategies:** Applying prevention, reduction, transfer, and acceptance techniques to manage vulnerabilities within AI/ML and data-driven systems.
- **Insurance Planning:** Selecting appropriate insurance products such as Cyber Liability Insurance, Technology Errors & Omissions (Tech E&O), Professional Indemnity, and Data Breach Insurance to transfer financial exposure.
- **Regulatory and Ethical Compliance:** Ensuring alignment with data protection laws (such as GDPR), algorithmic fairness guidelines, and cybersecurity standards.

VI. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and report preparation	1 per team
2	Internet access	For background study and references	As required
3	Reference materials	Articles, interviews, company profiles	1 set
4	Presentation slides / chart paper	For summary and display	1 per team

VIII. Safety Precautions

- a) **Ensure** all risk-related data verifiable sources.
- b) **Avoid** misrepresentation of risks or insurance coverage.

IX. Procedure

1. **Select** the CSE/AIML/DSCS/DS enterprise and clearly define its core product, users, data pipelines, and technology infrastructure.
2. **Identify** all potential risks related to cybersecurity, data breaches, AI/ML model failures, cloud outages, compliance issues, intellectual property, and financial or operational disruptions.
3. **Categorize** the identified risks into appropriate groups such as technical, operational, financial, legal/compliance, strategic, and external/environmental.
4. **Evaluate** the likelihood and severity of each risk using analytical tools such as risk matrices, heat maps, or probability–impact scoring.
5. **Examine** existing controls, safety mechanisms, encryption standards, backup systems, and compliance measures already implemented in the enterprise.
6. **Determine** appropriate mitigation strategies (avoid, reduce, transfer, accept) and map them to suitable insurance options like Cyber Liability Insurance, Tech E&O, Professional Indemnity, Data Breach Insurance, or Business Interruption Insurance.
7. **Prepare** a structured Risk & Insurance Plan summarizing the major risks, their impact, mitigation strategies, and recommended insurance policies.

8. **Review** the completed plan for accuracy, completeness, ethical correctness, confidentiality, and alignment with industry risk-management standards before submission.

X. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	Intel	Intel Core i5 processor with RTX 4050 GPU used for preparing the risk analysis report	1	N/A
2.	Internet Access	Wi-Fi	Used for researching risk management methods and insurance policies	1	N/A
3.	Browser	Chrome	Used to access online templates and reference materials	1	N/A
4.	Templates	Miro	Risk Matrix / Risk Heat Map template used for risk analysis	1	N/A

XI. Actual Procedure Followed

1. Selected the startup idea **Digital Attendance and Productivity Tracker** based on the mini-project concept.
2. Identified the main features of the system such as **automated attendance recording, productivity monitoring, and report generation for students and teachers**.
3. Identified different types of risks related to the project, including **technical risks, security risks, financial risks, operational risks, and legal/compliance risks**.
4. Categorized the risks into **internal and external risks** and prioritized them as **high, medium, or low** based on their importance.
5. Evaluated each risk using a **Risk Matrix approach**, analyzing the likelihood of occurrence and its potential impact on the startup.
6. Developed suitable **risk mitigation strategies** such as avoiding risks, reducing risks, transferring risks through insurance, or accepting minor risks.
7. Identified appropriate **insurance options** including cybersecurity insurance, data breach insurance, professional liability insurance, and business interruption insurance.
8. Prepared a **Risk & Insurance Plan** outlining the identified risks, their potential impact, mitigation strategies, and suitable insurance coverage.

XII. Observations and Data

S. No.	Entrepreneur	Major Venture	Key Traits	Opportunity Identified
1	Technical Risk	System failure or server downtime	High	Regular system monitoring and backup systems
2	Security Risk	Data breach or unauthorized access	High	Data encryption and secure authentication

S. No.	Entrepreneur	Major Venture	Key Traits	Opportunity Identified
3	Financial Risk	Development or operational cost increase	Medium	Proper financial planning and budgeting
4	Operational Risk	Poor team coordination or implementation issues	Medium	Structured project management and role allocation
5	Legal Risk	Data privacy and compliance issues	Medium	Compliance with data protection regulations

XIII. Results/Observations

1. The Digital Attendance and Productivity Tracker provides an effective digital solution for managing attendance and monitoring academic productivity in educational institutions.
2. The risk analysis revealed that technical risks and cybersecurity risks are the most critical issues that require careful management.
3. The study also showed that system reliability and data protection are essential for gaining user trust.
4. Risk mitigation strategies such as secure authentication, regular updates, system monitoring, and backup mechanisms can reduce the probability of system failure.
5. Insurance planning highlighted the importance of cybersecurity and data breach insurance in protecting the startup from unexpected financial losses.

XIV. Interpretation of Results

The analysis indicates that the **Digital Attendance and Productivity Tracker** is a practical startup idea, but it must address potential technical and security risks to ensure reliable operation. Implementing risk management strategies such as strong cybersecurity measures, regular system maintenance, and proper compliance with data protection regulations can significantly reduce operational risks.

In addition, insurance policies act as a **financial safety mechanism** that protects the business from unexpected incidents such as cyberattacks or service disruptions. Therefore, combining **risk mitigation strategies with insurance planning** strengthens the sustainability of the startup.

XV. Conclusions

The **Digital Attendance and Productivity Tracker** offers a useful solution for improving attendance management and productivity monitoring in educational institutions. This practicum helped identify major risks such as technical failures, cybersecurity threats, financial uncertainties, and operational challenges.

By implementing appropriate **risk management strategies and selecting suitable insurance policies**, the startup can minimize potential threats and ensure stable system performance. Effective risk and insurance planning plays an important role in building a **secure, reliable, and sustainable technology startup**.

XVI. Suggested Practicum Related Questions *(Every batch will be given another set of questions)*

1. **Evaluate** the major cybersecurity threats that could disrupt the operations of the selected CSE/AIML/DSCS/DS enterprise and propose evidence-based mitigation strategies.
2. **Identify** the most critical data privacy risks associated with the enterprise's product/service and justify the insurance coverage required to protect against financial loss.

3. **Analyze** the impact of algorithmic errors or model failures on the enterprise's business continuity and recommend corrective risk-management measures.
4. **Assess** the legal and regulatory risks (such as GDPR, IT Act, AI compliance guidelines) that may affect the enterprise and propose compliance mechanisms.
5. **Classify** the risks faced by the enterprise using a risk matrix and explain which risks should be prioritized for mitigation or insurance planning.
6. **Examine** the financial implications of a potential cyberattack or data breach and suggest suitable insurance products to transfer financial exposure.
7. **Critique** the enterprise's existing risk-control measures and identify gaps that may expose the startup to operational or reputational damage.
8. **Recommend** an incident-response plan for handling emergencies such as system downtime, cloud failures, or unauthorized data access.
9. **Predict** future risks emerging from evolving technologies such as generative AI, quantum computing, or autonomous systems that may affect the enterprise.
10. **Formulate** a comprehensive risk-insurance plan integrating risk identification, mitigation strategies, and appropriate insurance coverages for long-term sustainability.

XVII. References / Suggestions for further reading

Dorf, R.C., & Byers, T.H. *Technology Ventures: From Idea to Enterprise*. McGraw-Hill Education.

XVIII. Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the **process-related skills** and **product-related skills** will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Selection of relevant entrepreneur and data collection	3	
2	Identification and analysis of entrepreneurial traits	4	
3	Logical correlation with business opportunities	4	
4	Team collaboration and communication	3	
5	Ethical use of data and originality of report	4	
Product-related Skills: 12 Marks - 40%			
1	Clarity and structure of report presentation	3	
2	Depth of analysis and interpretation	3	
3	Accuracy of business opportunity linkage	3	
4	Quality of conclusions and insights	3	
5	Timely submission of report and presentation	3	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Signature with Date
Process-Related skills (18)	Product-Related skills (12)	Total (30)		

--	--	--	--	--

